# Derivatives and Taylor Series

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## Derivatives

### Limits and derivatives

**Definition 0.1** (Limit of a function at a point)**.** Let $f$ be a function defined at every point of an open interval around $c$, except possibly at $c$ itself, and let $L$ be a real number. The **limit** of $f(x)$ as $x$ approaches $c$ is $L$, written $\lim_{x \to c} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{for every } x \text{ with } 0 < |x - c| < \delta.$$

Here $|\cdot|$ is the absolute value. When such a real number $L$ exists, the limit **exists**; otherwise, the limit does not exist. The value $f(c)$, if it is defined, plays no role.

**Example 0.1** (The limit of $3x + 1$ at 2)**.** $\lim_{x \to 2}(3x + 1) = 7$. Given $\varepsilon > 0$, take $\delta = \varepsilon/3$. For every $x$ with $0 < |x - 2| < \delta$, using the distributive law in the second line,

$$\begin{aligned} |(3x + 1) - 7| &= |3x - 6| && \text{(subtract)} \\ &= |3(x - 2)| && \text{(distributive law)} \\ &= 3|x - 2| && (|3y| = 3|y|, \text{ since } 3 > 0) \\ &< 3 \cdot \frac{\varepsilon}{3} && (|x - 2| < \delta = \varepsilon/3) \\ &= \varepsilon && \text{(multiply)} \end{aligned}$$

For example, with $\varepsilon = 0.3$ and $\delta = 0.1$, the point $x = 2.05$ satisfies $0 < |2.05 - 2| < 0.1$, and

$$\begin{aligned} |(3 \cdot 2.05 + 1) - 7| &= |7.15 - 7| \\ &= 0.15 \\ &< 0.3. \end{aligned}$$

**Definition 0.2** (Right-hand limit)**.** Let $f$ be a function and $L$ a real number. The **right-hand limit** of $f$ at $c$ is $L$, written $\lim_{x \to c^+} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for every $x$ with $c < x < c + \delta$.

**Definition 0.3** (Left-hand limit)**.** Let $f$ be a function and $L$ a real number. The **left-hand limit** of $f$ at $c$ is $L$, written $\lim_{x \to c^-} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for every $x$ with $c - \delta < x < c$.

**Definition 0.4** (One-sided limits)**.** Let $f$ be a function. The right-hand limit (Definition 0.2) and the left-hand limit (Definition 0.3) of $f$ at $c$ are the **one-sided limits** of $f$ at $c$.

**Example 0.2** (One-sided limits of a step function)**.** Let $H(x) = 1$ for $x \geq 0$ and $H(x) = 0$ for $x < 0$.

- $\lim_{x \to 0^+} H(x) = 1$, because $H(x) = 1$ for every $x > 0$.
- $\lim_{x \to 0^-} H(x) = 0$, because $H(x) = 0$ for every $x < 0$.

The one-sided limits differ, since $1 \neq 0$, so $\lim_{x \to 0} H(x)$ does not exist.

Remark (A limit exists exactly when both one-sided limits exist and are equal)

*Remark* 0.1 (A limit exists exactly when both one-sided limits exist and are equal)*.* The limit $\lim_{x \to c} f(x)$ (Definition 0.1) exists exactly when both one-sided limits (Definition 0.4) exist and are equal, and then all three are equal.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist introduces limits and derivatives:

- Calculating Limits
- Exercises on Limits
- Intro to Differential Calculus
- How Derivatives Arise from Limits
- Derivative Notation

**Definition 0.5** (Difference quotient)**.** Let $f$ be a real-valued function of one variable, defined at $c$ and at $c + h$, with $h \neq 0$. The **difference quotient** of $f$ at $c$ with increment

$h$ is

$$\frac{f(c+h)-f(c)}{h},$$

the slope of the line through the points $(c, f(c))$ and $(c+h, f(c+h))$.

**Example 0.3** (The difference quotient of $x^2$ at 1)**.** For $f(x) = x^2$, $c = 1$, and $h \neq 0$,

$$\begin{aligned}
\frac{f(1+h)-f(1)}{h} &= \frac{(1+h)^2-1}{h} && \text{(substitute into } f\text{)} \\
&= \frac{1+2h+h^2-1}{h} && \text{(expand } (1+h)^2\text{)} \\
&= \frac{2h+h^2}{h} && (1-1=0) \\
&= 2+h, && \text{(divide by } h \neq 0\text{)}
\end{aligned}$$

which tends to 2 as $h \to 0$ (Definition 0.1). With $h = 0.1$ the difference quotient is $2 + 0.1 = 2.1$: the line through $(1, 1)$ and $(1.1, 1.21)$ has slope

$$\begin{aligned}
\tfrac{1.21-1}{0.1} &= \tfrac{0.21}{0.1} \\
&= 2.1.
\end{aligned}$$

**Definition 0.6** (Differentiable function)**.** A function $f$ is **differentiable at** $x = c$ if the limit (Definition 0.1) of the difference quotient (Definition 0.5) as $h \to 0$,

$$f'(c) \stackrel{\text{def}}{=} \lim_{h \to 0} \frac{f(c+h)-f(c)}{h},$$

exists and is finite. The number $f'(c)$ is the derivative of $f$ at $c$ (Definition 0.7). (Larson and Edwards 2018, sec. 2.1, p. 100)

**Example 0.4** (A differentiable function, and a limit that is not finite)**.**

- $f(x) = x^2$ is differentiable at $c = 3$:

$$\begin{aligned}
\frac{f(3+h)-f(3)}{h} &= \frac{(3+h)^2-3^2}{h} && \text{(substitute } f(x)=x^2\text{)}\\
&= \frac{9+6h+h^2-9}{h} && \text{(expand } (3+h)^2\text{)}\\
&= \frac{6h+h^2}{h} && \text{(cancel } 9-9\text{)}\\
&= 6+h && \text{(divide by } h\neq 0\text{)}
\end{aligned}$$

which tends to the finite limit $f'(3) = 6$ as $h \to 0$.

- $g(x) = \sqrt[3]{x}$ is not differentiable at $c = 0$: the difference quotient

$$\begin{aligned}
\frac{g(h)-g(0)}{h} &= \frac{\sqrt[3]{h}}{h}\\
&= \frac{1}{(\sqrt[3]{h})^2}
\end{aligned}$$

grows without bound as $h \to 0$, so the limit is not finite.

**Definition 0.7** (Derivative)**.** If $f$ is differentiable at $c$ (Definition 0.6), the number $f'(c)$ is the **derivative** of $f$ at $c$. The **derivative** of $f$ is the function $f'$ whose value at each $x$ where $f$ is differentiable is $f'(x)$.

**Definition 0.8** (Lagrange's notation)**.** Writing the derivative (Definition 0.7) with a prime, as $f'$ or $f'(x)$, is **Lagrange's notation**.

**Definition 0.9** (Leibniz notation)**.** The derivative of $f$ (Definition 0.7) is also written $\frac{df}{dx}$ or $\frac{d}{dx}f(x)$, which is **Leibniz notation**.

**Example 0.5** (The derivative of $x^2$)**.** For $f(x) = x^2$, the computation in Example 0.4, with 3 replaced by any real number $c$, gives $\frac{f(c+h)-f(c)}{h} = 2c + h$, which tends to $2c$ as $h \to 0$. So the derivative of $f$ is $f'(x) = 2x$. For example, $f'(3) = 6$ and $f'(-1) = -2$.

**Definition 0.10** (Right-hand derivative)**.** Let $f$ be a function defined at $c$. The **right-hand derivative** of $f$ at $c$ is the right-hand limit (Definition 0.2) $\lim_{h\to 0^+} \frac{f(c+h)-f(c)}{h}$ of the difference quotient (Definition 0.5), when that limit exists and is finite.

**Definition 0.11** (Left-hand derivative)**.** Let $f$ be a function defined at $c$. The **left-hand derivative** of $f$ at $c$ is the left-hand limit (Definition 0.3) $\lim_{h\to 0^-} \frac{f(c+h)-f(c)}{h}$, when that

limit exists and is finite.

**Definition 0.12** (One-sided derivatives)**.** Let $f$ be a function defined at $c$. The right-hand derivative (Definition 0.10) and the left-hand derivative (Definition 0.11) of $f$ at $c$ are the **one-sided derivatives** of $f$ at $c$.

**Example 0.6** (One-sided derivatives of $|x|$ at 0)**.** Let $f(x) = |x|$, the absolute value, and $c = 0$.

- For $h > 0$, $|h| = h$, so

$$\begin{aligned}\frac{f(0+h) - f(0)}{h} &= \frac{h - 0}{h} \\ &= 1.\end{aligned}$$

  The right-hand derivative of $f$ at 0 is 1.

- For $h < 0$, $|h| = -h$, so

$$\begin{aligned}\frac{f(0+h) - f(0)}{h} &= \frac{-h - 0}{h} \\ &= -1.\end{aligned}$$

  The left-hand derivative of $f$ at 0 is $-1$.

For example, $h = 0.5$ gives $\frac{0.5}{0.5} = 1$, and $h = -0.5$ gives $\frac{0.5}{-0.5} = -1$. The one-sided derivatives differ, since $1 \neq -1$.

**Definition 0.13** (Second derivative)**.** Let $f$ be a real-valued function of one variable whose derivative $f'$ (Definition 0.7) exists at every point of an open interval containing $c$. If $f'$ is differentiable (Definition 0.6) at $c$, its derivative there, written $f''(c)$ or $\frac{d^2 f}{dx^2}$, is the **second derivative** of $f$ at $c$.

**Example 0.7** (The second derivative of $x^3$)**.** For $f(x) = x^3$, the power rule (Theorem 0.3) gives $f'(x) = 3x^2$ at every $x$, and differentiating again, with the constant multiple rule (Theorem 0.2), gives

$$\begin{aligned}f''(x) &= 3 \cdot 2x \\ &= 6x.\end{aligned}$$

So $f''(2) = 12$, $f''(0) = 0$, and $f''(-1) = -6$.

### Derivative rules

**Theorem 0.1** (Constant rule)**.** *If $c$ is constant with respect to $x$, then*

$$\frac{\partial}{\partial x} c = 0$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Derivative of a Constant

**Theorem 0.2** (Constant multiple rule)**.** *If $a$ is constant with respect to $x$ and $y$ is a differentiable function of $x$, then:*

$$\frac{\partial}{\partial x} ay = a\frac{\partial y}{\partial x}$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Constant Multiple Rule for Derivatives

**Theorem 0.3** (Power rule)**.** *For every real number $q$ and every $x > 0$, the derivative of the power $x^q$ is:*

$$\frac{\partial}{\partial x} x^q = qx^{q-1}$$

*When $q$ is a positive integer, the same formula holds for every real $x$.*

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on this rule, on the sum rule (the derivative of a sum is the sum of the derivatives), and on exercises that combine the rules so far:

- The Power Rule for Derivatives
- The Sum Rule for Derivatives
- Exercises on Derivative Rules

**Exercise 0.1** (Derivative of polynomial and power functions)**.** Adapted from Miller (2016), Question 1.1.1.
Find the derivative of

$$f(x) = 4x^5 + 3x^2 + x^{1/3}$$

Solution

*Solution* 0.1. Differentiate term by term using the constant multiple rule (Theorem 0.2) and the power rule (Theorem 0.3):

$$\begin{aligned}
\frac{d}{dx}(4x^5) &= 4 \cdot (5x^4) \\
&= 20x^4 \\
\frac{d}{dx}(3x^2) &= 3 \cdot (2x) \\
&= 6x \\
\frac{d}{dx}(x^{1/3}) &= \frac{1}{3}x^{1/3-1} \\
&= \frac{1}{3}x^{-2/3}
\end{aligned}$$

Combining the three terms yields:

$$f'(x) = 20x^4 + 6x + \frac{1}{3}x^{-2/3}$$

**Exercise 0.2** (Finding points with a prescribed slope)**.** Adapted from Miller (2016), Question 1.1.14.
Let $f(x) = \frac{1}{3}x^3 + x^2 - x - 1$. Find all points on the graph of $f$ where the slope is:

1. $-1$
2. $2$
3. $0$

Solution

*Solution* 0.2. The slope is given by the derivative:

$$f'(x) = x^2 + 2x - 1$$

**1. Slope $-1$:** Solve

$$\begin{aligned} x^2 + 2x - 1 = -1 &\iff x^2 + 2x = 0 \\ &\iff x(x+2) = 0. \end{aligned}$$

The roots are $x = 0$ and $x = -2$. Evaluating $f$: $f(0) = -1$ and

$$\begin{aligned} f(-2) &= -\frac{8}{3} + 4 + 2 - 1 \\ &= \frac{7}{3}. \end{aligned}$$

The points are $(0, -1)$ and $(-2, 7/3)$.

**2. Slope 2:** Solve

$$\begin{aligned} x^2 + 2x - 1 = 2 &\iff x^2 + 2x - 3 = 0 \\ &\iff (x+3)(x-1) = 0. \end{aligned}$$

The roots are $x = -3$ and $x = 1$. Evaluating $f$:

$$\begin{aligned} f(-3) &= -9 + 9 + 3 - 1 \\ &= 2 \end{aligned}$$

and

$$\begin{aligned} f(1) &= \frac{1}{3} + 1 - 1 - 1 \\ &= -\frac{2}{3}. \end{aligned}$$

The points are $(-3, 2)$ and $(1, -2/3)$.

**3. Slope 0:** Solve $x^2 + 2x - 1 = 0$. By the quadratic formula:

$$\begin{aligned} x &= \frac{-2 \pm \sqrt{4 - 4(1)(-1)}}{2} \\ &= \frac{-2 \pm \sqrt{8}}{2} \\ &= -1 \pm \sqrt{2} \end{aligned}$$

**Theorem 0.4** (Derivative of natural logarithm)**.** *For every $x > 0$, the derivative of the natural logarithm is:*

$$\begin{aligned}\log'\{x\} &= \frac{1}{x} \\ &= x^{-1}\end{aligned}$$

**Exercise 0.3** (Derivative after logarithmic simplification)**.** Adapted from Miller (2016), Question 1.1.4.
Find the derivative of

$$f(x) = \log(4x) - \log(2x)$$

for $x > 0$.

Solution

*Solution* 0.3. Simplify before differentiating using the quotient property of logarithms:

$$\begin{aligned}f(x) &= \log(4x) - \log(2x) \\ &= \log\left(\frac{4x}{2x}\right) \\ &= \log 2\end{aligned}$$

Because $\log 2$ is constant with respect to $x$, the constant rule (Theorem 0.1) gives:

$$f'(x) = 0$$

*Remark:* While one could apply the chain rule separately to each term, giving

$$\begin{aligned}\frac{4}{4x} - \frac{2}{2x} &= \frac{1}{x} - \frac{1}{x} \\ &= 0,\end{aligned}$$

simplifying algebraically first is faster and prevents arithmetic errors.

**Theorem 0.5** (Derivative of exponential)**.** *For every real $x$, the derivative of the exponential function is:*

$$\exp'\{x\} = \exp\{x\}$$

**Theorem 0.6** (Derivative of sine)**.** *For every real $x$, the derivative of the sine function is:*

$$\frac{\partial}{\partial x} \sin x = \cos x$$

**Theorem 0.7** (Derivative of cosine)**.** *For every real $x$, the derivative of the cosine function is:*

$$\frac{\partial}{\partial x} \cos x = -\sin x$$

**Theorem 0.8** (Product rule)**.** *If $a$ and $b$ are differentiable functions of $x$, then*

$$(ab)' = ab' + ba'$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Product Rule for Derivatives

**Exercise 0.4** (Derivative of a product)**.** Adapted from Miller (2016), Question 1.1.2. Find the derivative of

$$f(x) = (x^4 + 3x^2 + 8) \cos x$$

Solution

*Solution* 0.4. Apply the product rule (Theorem 0.8) with:

$$u(x) = x^4 + 3x^2 + 8, \qquad v(x) = \cos x$$

Their derivatives are:

$$u'(x) = 4x^3 + 6x, \qquad v'(x) = -\sin x$$

By the product rule:

$$\begin{aligned} f'(x) &= u'(x)v(x) + u(x)v'(x) \\ &= (4x^3 + 6x) \cos x - (x^4 + 3x^2 + 8) \sin x \end{aligned}$$

**Exercise 0.5** (Second derivative of a product)**.** Adapted from Miller (2016), Question 1.1.15.
Find the second derivative of

$$f(x) = (x^4 + 3x^2 + 8)\cos x$$

Solution

*Solution* 0.5. From Exercise 0.4, the first derivative is:

$$f'(x) = (4x^3 + 6x)\cos x - (x^4 + 3x^2 + 8)\sin x$$

Differentiate each term via the product rule:

$$\frac{d}{dx}\left[(4x^3 + 6x)\cos x\right] = (12x^2 + 6)\cos x - (4x^3 + 6x)\sin x$$
$$\frac{d}{dx}\left[-(x^4 + 3x^2 + 8)\sin x\right] = -(4x^3 + 6x)\sin x - (x^4 + 3x^2 + 8)\cos x$$

Combining and grouping like trigonometric terms:

$$\begin{aligned} f''(x) &= \left[(12x^2 + 6) - (x^4 + 3x^2 + 8)\right]\cos x - 2(4x^3 + 6x)\sin x \\ &= (-x^4 + 9x^2 - 2)\cos x - (8x^3 + 12x)\sin x \end{aligned}$$

**Theorem 0.9** (Quotient rule)**.** *If $a$ and $b$ are differentiable functions of $x$ and $b \neq 0$, then*

$$(a/b)' = a'/b - (a/b^2)b'$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Quotient Rule for Derivatives

**Exercise 0.6** (Derivative of a rational function)**.** Adapted from Miller (2016), Question 1.1.12.
Find the derivative of

$$f(x) = \frac{x^2 - 1}{x - 1}$$

for $x \neq 1$.

Solution

*Solution* 0.6. Factor the numerator before differentiating:

$$\begin{aligned} f(x) &= \frac{(x-1)(x+1)}{x-1} \\ &= x+1 \qquad (x \neq 1) \end{aligned}$$

Differentiating directly gives $f'(x) = 1$.
*Alternative (quotient rule):* Using Theorem 0.9 with $u(x) = x^2 - 1$ and $v(x) = x - 1$:

$$\begin{aligned} f'(x) &= \frac{2x(x-1) - (x^2-1)(1)}{(x-1)^2} \\ &= \frac{x^2 - 2x + 1}{(x-1)^2} \\ &= \frac{(x-1)^2}{(x-1)^2} \\ &= 1 \end{aligned}$$

Both methods yield 1, but factoring first eliminates tedious algebraic simplification.

**Theorem 0.10** (Chain rule)**.** *If $a$ is a differentiable function of $b$, and $b$ is a differentiable function of $c$, then $a$ is a differentiable function of $c$, and*

$$\begin{aligned} \frac{da}{dc} &= \frac{da}{db}\frac{db}{dc} \\ &= \frac{db}{dc}\frac{da}{db} \end{aligned}$$

*or in Lagrange's notation (Definition 0.7), if $g$ is differentiable at $x$ and $f$ is differentiable at $g(x)$, then for the composition $f \circ g$, with inner function $g$ and outer function $f$:*

$$(f(g(x)))' = g'(x)f'(g(x))$$

**Exercise 0.7** (Derivative of the standard Gaussian kernel)**.** Adapted from Miller (2016), Question 1.1.5.
Find the derivative of

$$f(x) = \exp\left\{-\frac{x^2}{2}\right\}$$

Solution

*Solution* 0.7. Apply the chain rule (Theorem 0.10) with outer function $\exp\{u\}$ and inner function $u(x) = -x^2/2$. Since $\frac{d}{du}\exp\{u\} = \exp\{u\}$ (Theorem 0.5) and $u'(x) = -x$:

$$\begin{aligned} f'(x) &= u'(x)\exp\{u(x)\} \\ &= -x\exp\left\{-\frac{x^2}{2}\right\} \end{aligned}$$

**Theorem 0.11** (Derivative of the standard Gaussian kernel)**.** *For* $f(x) = \exp\left\{-\frac{x^2}{2}\right\}$ *and every* $x \in \mathbb{R}$*,*

$$f'(x) = -x\exp\left\{-\frac{x^2}{2}\right\}$$

Proof

*Proof.* This is the solution to Exercise 0.7, worked in Solution 0.7. □

**Exercise 0.8** (Second derivative of the Gaussian kernel)**.** Adapted from Miller (2016), Question 1.1.6.
Find the second derivative of

$$f(x) = \exp\left\{-\frac{x^2}{2}\right\}$$

Solution

*Solution* 0.8. From Theorem 0.11, the first derivative is $f'(x) = -x\exp\{-x^2/2\}$. Apply the product rule (Theorem 0.8) to $u(x) = -x$ and $v(x) = \exp\{-x^2/2\}$:

$$\begin{aligned} u'(x) &= -1 \\ v'(x) &= -x\exp\left\{-\frac{x^2}{2}\right\} \end{aligned}$$

Therefore:

$$
\begin{aligned}
f''(x) &= u'(x)v(x) + u(x)v'(x) \\
&= (-1)\exp\left\{-\frac{x^2}{2}\right\} + (-x)\left[-x\exp\left\{-\frac{x^2}{2}\right\}\right] \\
&= (x^2-1)\exp\left\{-\frac{x^2}{2}\right\}
\end{aligned}
$$

**Theorem 0.12** (Second derivative of the standard Gaussian kernel)**.** *For* $f(x) = \exp\left\{-\frac{x^2}{2}\right\}$ *and every* $x \in \mathbb{R}$,

$$
f''(x) = (x^2-1)\exp\left\{-\frac{x^2}{2}\right\}
$$

Proof

*Proof.* This is the solution to Exercise 0.8, worked in Solution 0.8. □

**Exercise 0.9** (Combining product and chain rules)**.** Adapted from Miller (2016), Question 1.1.7.
Find the derivative of

$$
f(x) = \exp\{x^8\}\cos(3x^4)
$$

Solution

*Solution* 0.9. Apply the product rule (Theorem 0.8) to $u(x) = \exp\{x^8\}$ and $v(x) = \cos(3x^4)$:

$$
f'(x) = u'(x)v(x) + u(x)v'(x)
$$

Compute $u'(x)$ and $v'(x)$ via the chain rule (Theorem 0.10):

$$
\begin{aligned}
u'(x) &= 8x^7\exp\{x^8\} \\
v'(x) &= -12x^3\sin(3x^4)
\end{aligned}
$$

Substitute both derivatives into the product rule:

$$
\begin{aligned}
f'(x) &= 8x^7\exp\{x^8\}\cos(3x^4) - 12x^3\exp\{x^8\}\sin(3x^4) \\
&= 4x^3\exp\{x^8\}[2x^4\cos(3x^4) - 3\sin(3x^4)]
\end{aligned}
$$

**Exercise 0.10** (Generalized power rule)**.** Adapted from Miller (2016), Question 1.1.13. Find the derivative of

$$\begin{aligned} f(x) &= \sqrt[3]{(5x-2)^2} \\ &= (5x-2)^{2/3} \end{aligned}$$

for $x \neq 2/5$.

Solution

*Solution* 0.10. Apply the generalized power rule $\frac{d}{dx}[g(x)]^r = r[g(x)]^{r-1}g'(x)$ with $g(x) = 5x - 2$ and $r = 2/3$:

$$\begin{aligned} f'(x) &= \frac{2}{3}(5x-2)^{2/3-1} \cdot 5 \\ &= \frac{10}{3}(5x-2)^{-1/3} \\ &= \frac{10}{3(5x-2)^{1/3}} \end{aligned}$$

*Remark:* Omitting the inner derivative $g'(x) = 5$ is a frequent mistake when applying the generalized power rule.

**Corollary 0.1** (Chain rule for logarithms)**.** *If $f$ is differentiable at $x$ and $f(x) > 0$, then*

$$\frac{d}{dx}\log\{f(x)\} = \frac{f'(x)}{f(x)}$$

Proof

*Proof.* Apply Theorem 0.10 and Theorem 0.4:

$$\begin{aligned} \frac{d}{dx}\log\{f(x)\} &= f'(x) \cdot \log'\{f(x)\} && \text{(chain rule, with } g = f \text{ and outer function } \log\text{)} \\ &= f'(x) \cdot \frac{1}{f(x)} && \text{(derivative of } \log\text{, valid because } f(x) > 0\text{)} \\ &= \frac{f'(x)}{f(x)} && \text{(multiply)} \end{aligned}$$

□

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on this rule and on exercises that combine it with the other rules:

- The Chain Rule for Derivatives
- The Power Rule on a Function Chain
- Advanced Exercises on Derivative Rules

**Exercise 0.11** (Chain rule with logarithm)**.** Adapted from Miller (2016), Question 1.1.3. Find the derivative of

$$f(x) = \log(1 - x^2)$$

for $x \in (-1, 1)$.

Solution

*Solution* 0.11. Apply the chain rule for logarithms (Corollary 0.1): if $g(x) = 1 - x^2$, then $g'(x) = -2x$, and

$$\begin{aligned} f'(x) &= \frac{g'(x)}{g(x)} \\ &= -\frac{2x}{1 - x^2} \end{aligned}$$

*Remark:* A common pitfall in chain rule problems is evaluating the outer derivative at $x$ rather than at the inner value $g(x)$. The denominator is $g(x) = 1 - x^2$, not $x$.

**Exercise 0.12** (Second derivative with an added constant)**.** Adapted from Miller (2016), Question 1.1.9.
Find the second derivative of

$$f(x) = \log x + \sqrt{162}$$

for $x > 0$.

Solution

*Solution* 0.12. Because $\sqrt{162}$ is constant with respect to $x$, its derivative is zero (Theorem 0.1):

$$\begin{aligned} f'(x) &= \frac{1}{x} \\ &= x^{-1} \end{aligned}$$

Differentiating again via the power rule (Theorem 0.3):

$$\begin{aligned} f''(x) &= -x^{-2} \\ &= -\frac{1}{x^2} \end{aligned}$$

### Studying a new function

**Definition 0.14** (Logarithmic derivative)**.** The **logarithmic derivative** $\frac{f'(x)}{f(x)}$ of a function $f$ of a real variable equals $\frac{d}{dx}\log\{f(x)\}$ by Corollary 0.1 and measures the rate of change of $f$ relative to its size; it needs $f(x) > 0$, and for $f(x) < 0$ the same expression is the derivative of $\log\{|f(x)|\}$.

**Definition 0.15** (Profile of a function)**.** The **profile** of a function $f$ of a real variable is the list of related functions and values to work out when meeting $f$ for the first time:

- the logarithm $\log\{f(x)\}$, where $f(x) > 0$;
- the derivatives $f'(x)$, $f''(x)$, and so on, as far as they are informative;
- the logarithmic derivative $\frac{f'(x)}{f(x)}$;
- the inverse function $f^{-1}$ (definition), which exists on the range of $f$ when $f$ is one-to-one, and otherwise only after restricting the domain of $f$;
- the complement $1 - f(x)$, which is informative when $f$ takes values in $[0, 1]$, as a probability does;
- the reciprocal $\frac{1}{f(x)}$, where $f(x) \neq 0$ (not to be confused with $f^{-1}$);
- an antiderivative of $f$, a function $F$ with $F' = f$ (integration);
- the values $f(0)$ and $f(1)$, and the limits of $f(x)$ as $x \to \infty$ and as $x \to -\infty$, at each of these that lies in the domain of $f$ or exists;
- the graph of $f$.

**Example 0.8** (The profile of the logistic function)**.** The **logistic function** (also called expit or the sigmoid) is

$$f(x) = \frac{1}{1 + e^{-x}}.$$

The profile in Definition 0.15 of $f$ is as follows. Write $E = e^{-x}$, so that $f = \frac{1}{1+E}$ and $1 - f = \frac{E}{1+E}$.

- **Logarithm:** $\log\{f(x)\} = -\log\{1 + e^{-x}\}$, defined for every real $x$ because $f(x) > 0$.
- **Derivative:** by the chain rule,

$$\begin{aligned} f'(x) &= \frac{e^{-x}}{(1+e^{-x})^2} && \text{(chain rule applied to } (1+e^{-x})^{-1}) \\ &= \frac{1}{1+E} \cdot \frac{E}{1+E} && \text{(split the square)} \\ &= f(x)\,(1 - f(x)). && \text{(identify } f \text{ and } 1-f) \end{aligned}$$

  Differentiating again with the product rule gives $f''(x) = f'(1-f) - ff' = f(x)(1-f(x))(1-2f(x))$.
- **Logarithmic derivative:** $\frac{f'(x)}{f(x)} = 1 - f(x)$, so the relative rate of change of $f$ is the complement of $f$ itself.
- **Inverse:** for $y \in (0,1)$, solving $y = \frac{1}{1+e^{-x}}$ for $x$ gives

$$\begin{aligned} \frac{1}{y} &= 1 + e^{-x} \\ e^{-x} &= \frac{1-y}{y} \\ x &= \log\left\{\frac{y}{1-y}\right\}. \end{aligned}$$

  So $f^{-1}(y) = \text{logit}\{y\}$, the log-odds. The inverse exists because $f' = f(1-f) > 0$, so $f$ is strictly increasing and therefore one-to-one.
- **Complement:** $1 - f(x) = \frac{e^{-x}}{1+e^{-x}} = \frac{1}{e^{x}+1} = f(-x)$.
- **Reciprocal:** $\frac{1}{f(x)} = 1 + e^{-x}$.
- **Antiderivative:** $F(x) = \log\{1 + e^{x}\}$ satisfies $F'(x) = \frac{e^{x}}{1+e^{x}} = f(x)$, so $\int f(x)\,dx = \log\{1 + e^{x}\} + C$.
- **Key values:** $f(0) = \frac{1}{1+1} = \frac{1}{2}$ and $f(1) = \frac{1}{1+e^{-1}}$. As $x \to \infty$, $e^{-x} \to 0$, so $f(x) \to 1$; as $x \to -\infty$, $e^{-x} \to \infty$, so $f(x) \to 0$.

The code checks these results numerically, using the base R functions `plogis()` for $f$ and `qlogis()` for $f^{-1}$, and `D()` to differentiate symbolically.

```
f <- plogis
x <- c(-2, -0.5, 0, 1, 3)

f_expr <- quote(1 / (1 + exp(-x)))
f_prime <- eval(D(f_expr, "x"), list(x = x))
f_second <- eval(D(D(f_expr, "x"), "x"), list(x = x))

checks <- c(
  derivative = max(abs(f_prime - f(x) * (1 - f(x)))),
  second_derivative = max(abs(f_second - f(x) * (1 - f(x)) * (1 - 2 * f(x)))),
  log_derivative = max(abs(f_prime / f(x) - (1 - f(x)))),
  inverse = max(abs(qlogis(f(x)) - x)),
  complement = max(abs(1 - f(x) - f(-x))),
  reciprocal = max(abs(1 / f(x) - (1 + exp(-x)))),
  antiderivative = abs(
    integrate(f, 0, 1)$value - (log(1 + exp(1)) - log(2))
  )
)
signif(checks, 3)
#>        derivative second_derivative    log_derivative           inverse
#>          1.46e-16          1.18e-16          1.53e-16          3.11e-15
#>        complement        reciprocal    antiderivative
#>          1.46e-16          0.00e+00          0.00e+00
stopifnot(all(checks < 1e-14))

key_values <- c(f(0), f(1), f(10), f(-10))
```

Each entry of `checks` is the largest discrepancy between a formula above and a numerical computation; every one is below $10^{-14}$ (the chunk stops with an error otherwise), which is rounding error. The key values are $f(0) = 0.5$ and $f(1) \approx 0.7311$. At $x = 10$ and $x = -10$ the function is already close to its limits, $f(10) \approx 0.99995$ and $f(-10) \approx 0.0000454$. Figure 1 graphs $f$, $f'$ and $\log\{f\}$.

```r
grid <- seq(-6, 6, length.out = 241)
panels <- data.frame(
  x = rep(grid, 3),
  panel = factor(
    rep(c("f(x)", "f'(x)", "log f(x)"), each = length(grid)),
    levels = c("f(x)", "f'(x)", "log f(x)")
  ),
  value = c(f(grid), f(grid) * (1 - f(grid)), log(f(grid)))
)
ggplot2::ggplot(panels) +
  ggplot2::aes(x = x, y = value) +
  ggplot2::geom_line() +
  ggplot2::facet_wrap(~panel, scales = "free_y") +
  ggplot2::labs(x = "x", y = NULL)
```

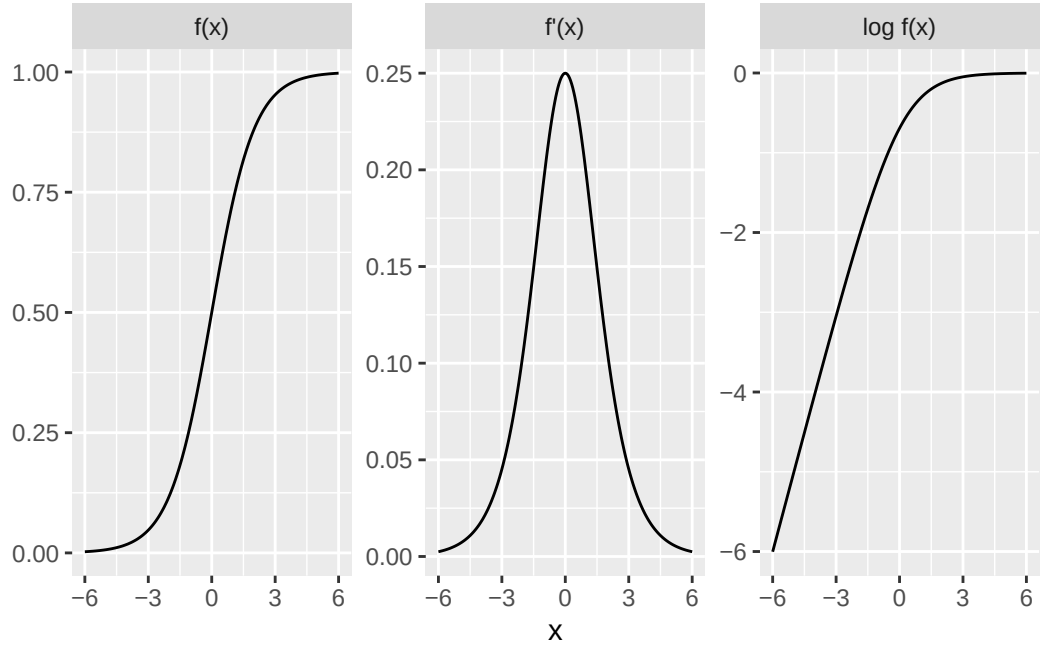



Figure 1: The logistic function, its derivative and its logarithm

### Tangent lines

**Definition 0.16** (Tangent line)**.** Let $f$ be differentiable at $c$ (Definition 0.6). The **tangent line** to the graph of $f$ at $c$ is the graph of the affine function

$$x \mapsto f(c) + f'(c)\,(x - c),$$

the line through the point $(c, f(c))$ with slope $f'(c)$. The slope $f'(c)$ is also called the **tangent slope** of $f$ at $c$.

**Example 0.9** (The tangent line to $x^2$ at 1)**.** For $f(x) = x^2$ at $c = 1$, $f(1) = 1$ and $f'(1) = 2$ (Example 0.5), so the tangent line is the graph of

$$\begin{aligned} x &\mapsto 1 + 2\,(x - 1) && \text{(substitute } f(1) = 1 \text{ and } f'(1) = 2\text{)} \\ &= 2x - 1 && \text{(distribute, and } 1 - 2 = -1\text{)} \end{aligned}$$

At $x = 1.1$ the tangent line has height $2(1.1) - 1 = 1.2$, close to the curve's height $f(1.1) = 1.21$; the gap, $1.21 - 1.2 = 0.01$, is $(1.1 - 1)^2$.

### Linear approximation

For a differentiable function $f$ and a small step $\varepsilon$,

$$f(w + \varepsilon) \approx f(w) + \varepsilon \frac{d}{dw} f(w) \tag{1}$$

**Definition 0.17** (Linear approximation (first-order Taylor approximation))**.** For a function $f$ that is differentiable at $w$, the **linear approximation** of $f$ at $w$ (also called the **first-order Taylor approximation**) is the function of the step $\varepsilon$

$$\hat{f}_w(\varepsilon) = f(w) + \varepsilon \frac{d}{dw} f(w)$$

so Equation 1 says $f(w + \varepsilon) \approx \hat{f}_w(\varepsilon)$ for small $\varepsilon$. Figure 2 illustrates this approximation interactively for a quadratic function.

ℹ Remark (The "linear" approximation is affine)

*Remark* 0.2 (The "linear" approximation is affine). The linear approximation $\hat{f}_w$ (Definition 0.17) is an affine function of the step $\varepsilon$: its slope is $\frac{d}{dw} f(w)$ and its intercept is $f(w)$.

It is linear in $\varepsilon$ only when $f(w) = 0$. The name "linear approximation" uses "linear" in the looser sense of elementary algebra (remark). Boyd and Vandenberghe (2018), section 2.2, treats the first-order Taylor approximation as an affine function for functions of several variables as well.

*(This plot is interactive; see the HTML or slide version of this page to view it.)*

Figure 2: The linear approximation Equation 1 for $f(w) = w^2 - 4w + 7$, at any $w$ and step $\varepsilon$.

**Exercise 0.13** (Tangent line approximation)**.** Adapted from Miller (2016), Question 1.1.8.
Let $f(x) = 4x + \sqrt{2}\cos x$.

1. Compute $f'(x)$.
2. Find the equation of the tangent line to the curve $y = f(x)$ at $x = \pi/4$.
3. Use the tangent line to approximate $f(\pi/4 + 0.01)$, and compare that approximation with the exact value.

Solution

*Solution* 0.13. **1.** Differentiate term by term:

$$f'(x) = 4 - \sqrt{2}\sin x$$

**2.** Evaluate $f$ and $f'$ at $x_0 = \pi/4$:

$$\begin{aligned} f(\pi/4) &= 4\left(\frac{\pi}{4}\right) + \sqrt{2}\cos\left(\frac{\pi}{4}\right) \\ &= \pi + \sqrt{2}\left(\frac{\sqrt{2}}{2}\right) \\ &= \pi + 1 \\ f'(\pi/4) &= 4 - \sqrt{2}\sin\left(\frac{\pi}{4}\right) \\ &= 4 - \sqrt{2}\left(\frac{\sqrt{2}}{2}\right) \\ &= 4 - 1 \\ &= 3 \end{aligned}$$

Using the point-slope formula, the tangent line at $(\pi/4, \pi + 1)$ is:

$$y - (\pi + 1) = 3\left(x - \frac{\pi}{4}\right) \implies y = \pi + 1 + 3\left(x - \frac{\pi}{4}\right)$$

**3.** At $x = \pi/4 + 0.01$, the linear approximation (Equation 1) gives:

$$f(\pi/4 + 0.01) \approx (\pi + 1) + 3(0.01) = \pi + 1.03 \approx 4.171593$$

The exact value is:

$$f(\pi/4 + 0.01) = 4\left(\frac{\pi}{4} + 0.01\right) + \sqrt{2}\cos\left(\frac{\pi}{4} + 0.01\right) \approx 4.171543$$

The approximation error is $|4.171593 - 4.171543| \approx 0.00005$, on the order of $(0.01)^2 = 10^{-4}$.

### Critical points and optimization

**Definition 0.18** (Flat point (stationary point))**.** Let $f$ be differentiable at $c$ (Definition 0.6). If $f'(c) = 0$, then $c$ is a **flat point** of $f$ (also called a **stationary point**).

**Example 0.10** (Flat points, and a point that is not one)**.**

- $f(w) = w^3$ has $f'(w) = 3w^2$, which is 0 at $w = 0$, so 0 is a flat point. Yet $f(0) = 0$ is neither the minimum nor the maximum of the values $f$ takes on any open interval around 0: $f(w) < 0$ for $w < 0$ and $f(w) > 0$ for $w > 0$.
- $h(w) = w^2$ has $h'(1) = 2 \neq 0$, so 1 is not a flat point.

**Exercise 0.14** (Find the flat point, and check the approximation)**.** Let $f(w) = w^2 - 4w + 7$.

1. Differentiate $f$.
2. Find the flat point $w$ of $f$, and say whether $f(w)$ is the minimum or the maximum of the values of $f$.
3. Evaluate the derivative at $w = 1$, use Equation 1 to predict $f(1.01)$, and compare that prediction with the exact value.

Solution

*Solution* 0.14. **1.** Term by term:

$$\frac{df}{dw} = 2w - 4$$

**2.** Set it to zero: $2w - 4 = 0$ gives $w = 2$.

$$\begin{aligned} f(2) &= 4 - 8 + 7 \\ &= 3 \end{aligned}$$

is the minimum of the values of $f$. Completing the square, $f(w) = (w-2)^2 + 3$, since

$$\begin{aligned} (w-2)^2 + 3 &= w^2 - 4w + 4 + 3 \\ &= w^2 - 4w + 7, \end{aligned}$$

and $(w-2)^2 \geq 0$, so $f(w) \geq 3 = f(2)$ for every $w$. Equivalently, the derivative is negative below $w = 2$ and positive above it, so the function falls into that point and rises out of it.
**3.** At $w = 1$ the derivative is $2(1) - 4 = -2$, so $f$ is falling there. With $\varepsilon = 0.01$, Equation 1 predicts a change of $(0.01)(-2) = -0.02$, from

$$\begin{aligned} f(1) &= 1 - 4 + 7 \\ &= 4 \end{aligned}$$

to 3.98. The exact value is

$$\begin{aligned} f(1.01) &= (1.01)^2 - 4(1.01) + 7 \\ &= 1.0201 - 4.04 + 7 \\ &= 3.9801 \end{aligned}$$

a change of $-0.0199$. The prediction is off by 0.0001, which is $\varepsilon^2$: the linear approximation drops everything of that order and smaller, so halving the step quarters the error. That trade is the whole bargain of gradient descent, the step-by-step method of fitting models defined on the optimization page. We take a step in the direction the derivative recommends, and the recommendation is trustworthy only as far as the step is small.

**Definition 0.19** (Critical point)**.** Let $f$ be a function defined on an open interval containing $c$. The point $c$ is a **critical point** of $f$ if either $f'(c) = 0$ or $f$ is not differentiable at $c$ (Definition 0.6).

**Example 0.11** (Critical points that are and are not flat points)**.**

- $g(w) = |w|$ has no derivative at $w = 0$: its one-sided derivatives there are 1 and $-1$ (Example 0.6), so the difference quotient $\frac{|h|-0}{h}$ has no limit as $h \to 0$ (Definition 0.4). So 0 is a critical point of $g$ but not a flat point.
- $f(w) = w^3$ has

$$\begin{aligned} f'(0) &= 3 \cdot 0^2 \\ &= 0, \end{aligned}$$

so 0 is a flat point of $f$, and hence a critical point.

- $h(w) = w^2$ is differentiable everywhere, with $h'(w) = 2w$, which is 0 only at $w = 0$. So 0 is the only critical point of $h$; for example, $h'(1) = 2 \neq 0$, so 1 is not one.

**Theorem 0.13** (Every flat point is a critical point)**.** *If $c$ is a flat point of $f$ (Definition 0.18), then $c$ is a critical point of $f$ (Definition 0.19).*

Proof

*Proof.* A flat point $c$ has $f$ differentiable at $c$ and $f'(c) = 0$. The condition "$f'(c) = 0$" is the first alternative in Definition 0.19, so $c$ is a critical point. □

**Example 0.12** (The flat points of a cubic, and a critical point that is not flat)**.** For $f(w) = w^3 - 3w$, the derivative is $f'(w) = 3w^2 - 3$. The code differentiates $f$ with `D()` and evaluates $f'$ at several points. For $g(w) = |w|$, it computes the difference quotients at 0 from both sides.

```
f_prime <- D(quote(w^3 - 3 * w), "w")
points <- c(-2, -1, 0, 1, 2)
derivs <- data.frame(w = points, f_prime = eval(f_prime, list(w = points)))
derivs
#>    w f_prime
#> 1 -2       9
#> 2 -1       0
#> 3  0      -3
#> 4  1       0
#> 5  2       9
flat <- derivs$w[derivs$f_prime == 0]

h <- c(-1e-3, 1e-3)
quotients <- abs(h) / h
quotients
#> [1] -1  1
```

The derivative is 0 at 2 of the 5 points, namely $w = -1$ and $w = 1$. These are flat points of $f$, so by Theorem 0.13 they are critical points. The difference quotients of $g$ at 0 are -1 from the left and 1 from the right, so $g$ has no derivative at 0. Thus 0 is a critical point of $g$ (Definition 0.19), but not a flat point.

**Exercise 0.15** (Maximizing a gamma-family kernel)**.** Adapted from Miller (2016), Question 1.1.10.
Find the maximum value of

$$f(x) = x^4 e^{-x}$$

on the interval $[0, \infty)$.

Solution

*Solution* 0.15. Differentiate using the product rule:

$$\begin{aligned} f'(x) &= 4x^3 e^{-x} - x^4 e^{-x} \\ &= x^3 e^{-x}(4 - x) \end{aligned}$$

On $[0, \infty)$, $e^{-x} > 0$, so $f'(x) = 0$ only at $x = 0$ and $x = 4$.
Evaluate $f$ at the boundary and critical points:

- At $x = 0$: $f(0) = 0$.
- At $x = 4$:

$$\begin{aligned} f(4) &= 4^4 e^{-4} \\ &= 256 e^{-4} \\ &\approx 4.6888. \end{aligned}$$

- As $x \to \infty$: exponential decay dominates polynomial growth, so $\lim_{x \to \infty} x^4 e^{-x} = 0$.

Since $f(x) \geq 0$ for all $x \geq 0$ and $f(4) > 0$, the global maximum occurs at $x = 4$, with maximum value $256/e^4$.

**Exercise 0.16** (Classifying critical points with the second derivative test)**.** Adapted from Miller (2016), Question 1.1.11.
Find the critical points of

$$f(x) = 4x^3 - 3x^2$$

and decide whether each is a local maximum, a local minimum, or an inflection point.

Solution

*Solution* 0.16. Compute the first derivative:

$$\begin{aligned} f'(x) &= 12x^2 - 6x \\ &= 6x(2x-1) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points at $x = 0$ and $x = 1/2$.
Compute the second derivative:

$$f''(x) = 24x - 6$$

Apply the second derivative test at each critical point:

- At $x = 0$: $f''(0) = -6 < 0$, so $x = 0$ is a strict local maximum.
- At $x = 1/2$:

$$\begin{aligned} f''(1/2) &= 24(1/2) - 6 \\ &= 6 \\ &> 0, \end{aligned}$$

  so $x = 1/2$ is a strict local minimum.

Inflection points occur where the second derivative changes sign ($f''(x) = 0$ at $x = 1/4$). Because $f''$ is non-zero at both critical points, neither is an inflection point.

### Taylor series

**Definition 0.20** (Taylor polynomial)**.** Let $f$ be a function that has at least $n$ derivatives at a point $x_0$. The **Taylor polynomial** of degree $n$ for $f$ centered at $x_0$ is:

$$\begin{aligned} P_n(x) &= \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \\ &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \end{aligned} \tag{2}$$

**Example 0.13** (Taylor series of the exponential function)**.** For $f(x) = e^x$ centered at $x_0 = 0$, every derivative is

$$\begin{aligned} f^{(k)}(0) &= e^0 \\ &= 1. \end{aligned}$$

The degree-$n$ Maclaurin polynomial is:

$$\begin{aligned} P_n(x) &= 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} \\ &= \sum_{k=0}^{n} \frac{x^k}{k!} \end{aligned}$$

Because the remainder $R_n(x) \to 0$ as $n \to \infty$ for all $x \in \mathbb{R}$, the Taylor series converges everywhere to $e^x$:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

**Definition 0.21** (Maclaurin polynomial)**.** When $x_0 = 0$, the Taylor polynomial $P_n(x)$ (Definition 0.20) is called the **Maclaurin polynomial**.

**Definition 0.22** (Taylor series)**.** Let $f$ and $x_0$ be as in Definition 0.20. When $f$ is infinitely differentiable and the series converges to $f(x)$ on an open interval containing $x_0$, the infinite sum $\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$ is the **Taylor series** of $f$ centered at $x_0$.

Video

Chapter 11 of the *Essence of calculus* series by 3Blue1Brown develops the geometric and polynomial intuition behind Taylor approximations:

- Taylor series shows how higher-order derivatives match the curvature and rate of change of a function near a point to build polynomial approximations (Definition 0.20).

**Exercise 0.17** (Taylor polynomial of a polynomial at the origin)**.** Adapted from Miller (2016), Question 1.1.16.
Find the first five terms (up to degree 4) of the Taylor series for

$$f(x) = x^8 + x^4 + 3$$

at $x_0 = 0$.

Solution

*Solution* 0.17. Evaluate $f$ and its first four derivatives at $x = 0$:

$$\begin{aligned} f(x) &= x^8 + x^4 + 3 && \implies f(0) = 3 \\ f'(x) &= 8x^7 + 4x^3 && \implies f'(0) = 0 \\ f''(x) &= 56x^6 + 12x^2 && \implies f''(0) = 0 \\ f'''(x) &= 336x^5 + 24x && \implies f'''(0) = 0 \\ f^{(4)}(x) &= 1680x^4 + 24 && \implies f^{(4)}(0) = 24 \end{aligned}$$

By Equation 2:

$$\begin{aligned} P_4(x) &= 3 + 0x + \frac{0}{2!}x^2 + \frac{0}{3!}x^3 + \frac{24}{4!}x^4 \\ &= 3 + x^4 \end{aligned}$$

Because $f$ is already a polynomial centered at 0, its Taylor polynomial of degree 4 simply collects its terms of degree at most 4.

**Exercise 0.18** (Taylor expansion of a polynomial at a nonzero point)**.** Adapted from Miller (2016), Question 1.1.17.
Find the first three terms (up to degree 2) of the Taylor series for

$$f(x) = x^8 + x^4 + 3$$

at $x_0 = 1$.

Solution

*Solution* 0.18. Evaluate $f$ and its first two derivatives at $x_0 = 1$:

$$\begin{aligned} f(1) &= 1^8 + 1^4 + 3 \\ &= 5 \\ f'(1) &= 8(1)^7 + 4(1)^3 \\ &= 12 \\ f''(1) &= 56(1)^6 + 12(1)^2 \\ &= 68 \end{aligned}$$

The degree-2 Taylor polynomial is:

$$\begin{aligned} P_2(x) &= f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 \\ &= 5 + 12(x-1) + 34(x-1)^2 \end{aligned}$$

**Example 0.14** (Taylor series of the cosine function)**.** For $f(x) = \cos x$ centered at $x_0 = 0$, the derivatives follow a repeating cycle of length 4:

$$\begin{aligned} f(0) &= \cos(0) \\ &= 1, \\ f'(0) &= -\sin(0) \\ &= 0, \\ f''(0) &= -\cos(0) \\ &= -1, \\ f'''(0) &= \sin(0) \\ &= 0, \end{aligned}$$

and in general $f^{(2k)}(0) = (-1)^k$ and $f^{(2k+1)}(0) = 0$ for all $k \geq 0$. The Taylor series (Maclaurin series) converges everywhere to $\cos x$:

$$\begin{aligned} \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + ... \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} \end{aligned}$$

**Example 0.15** (Taylor series of the sine function)**.** For $f(x) = \sin x$ centered at $x_0 = 0$, the derivatives evaluate at 0 to $f^{(2k)}(0) = 0$ and $f^{(2k+1)}(0) = (-1)^k$ for all $k \geq 0$. The Taylor series converges everywhere to $\sin x$:

$$\begin{aligned} \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + ... \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} \end{aligned}$$

**Exercise 0.19** (Taylor expansion of a scaled cosine)**.** Adapted from Miller (2016), Question 1.1.18.
Find the terms up to degree 4 in the Taylor series for

$$f(x) = \cos(5x)$$

at $x_0 = 0$.

Solution

*Solution* 0.19. Recall the standard Maclaurin series for cosine:

$$\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - ...$$

Substitute $u = 5x$:

$$\begin{aligned}\cos(5x) &= 1 - \frac{(5x)^2}{2!} + \frac{(5x)^4}{4!} - ... \\ &= 1 - \frac{25}{2}x^2 + \frac{625}{24}x^4 - ...\end{aligned}$$

*Remark:* Substituting into known Taylor expansions avoids computing high-order derivatives by hand.

**Exercise 0.20** (Taylor series of a power of a function)**.** Adapted from Miller (2016), Question 1.1.19.
Find the terms up to degree 4 in the Taylor series for

$$f(x) = \cos^3(5x)$$

at $x_0 = 0$.

Solution

*Solution* 0.20. From Exercise 0.19, the expansion of $\cos(5x)$ up to order $x^4$ is:

$$\cos(5x) = 1 - \frac{25}{2}x^2 + \frac{625}{24}x^4 + O(x^6)$$

Cube this expansion using $(1-y)^3 = 1 - 3y + 3y^2 - y^3$ with $y = \frac{25}{2}x^2 - \frac{625}{24}x^4$:

$$\begin{aligned}\cos^3(5x) &= 1 - 3\left(\frac{25}{2}x^2 - \frac{625}{24}x^4\right) + 3\left(\frac{25}{2}x^2\right)^2 + O(x^6) \\ &= 1 - \frac{75}{2}x^2 + \frac{625}{8}x^4 + \frac{1875}{4}x^4 + O(x^6) \\ &= 1 - \frac{75}{2}x^2 + \frac{4375}{8}x^4 + O(x^6)\end{aligned}$$

**Exercise 0.21** (Linear Taylor polynomial of the exponential)**.** Adapted from Miller (2016), Question 1.1.20.
Find the first two terms (degree 1) of the Taylor series for

$$f(x) = e^x$$

at $x_0 = 0$.

Solution

*Solution* 0.21. Since

$$\begin{aligned} f(0) &= e^0 \\ &= 1 \end{aligned}$$

and

$$\begin{aligned} f'(0) &= e^0 \\ &= 1 : \end{aligned}$$

$$\begin{aligned} P_1(x) &= f(0) + f'(0)x \\ &= 1 + x \end{aligned}$$

The full Taylor series is

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \\ &= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots . \end{aligned}$$

**Exercise 0.22** (Taylor series of a composite exponential)**.** Adapted from Miller (2016), Question 1.1.21.
Find the first six terms (up to degree 5) of the Taylor series for

$$f(x) = \exp\{x^8\}$$

at $x_0 = 0$.

Solution

*Solution* 0.22. Substitute $u = x^8$ into $e^u = 1 + u + \frac{u^2}{2!} + \dots$:

$$\exp\{x^8\} = 1 + x^8 + \frac{x^{16}}{2} + ...$$

Because the lowest-order variable term is $x^8$, all coefficients for powers $x^1, x^2, x^3, x^4, x^5$ are zero:

$$\begin{aligned} P_5(x) &= 1 + 0x + 0x^2 + 0x^3 + 0x^4 + 0x^5 \\ &= 1 \end{aligned}$$

**Exercise 0.23** (Taylor expansion of the normal density)**.** Adapted from Miller (2016), Question 1.1.22.
Find the terms up to degree 3 of the Taylor series for the standard normal density

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}$$

at $x_0 = 0$.

Solution

*Solution* 0.23. Using $e^u = 1 + u + O(u^2)$ with $u = -x^2/2$:

$$\exp\left\{-\frac{x^2}{2}\right\} = 1 - \frac{x^2}{2} + O(x^4)$$

Multiplying by the normalization constant $1/\sqrt{2\pi}$:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} - \frac{x^2}{2\sqrt{2\pi}} + O(x^4)$$

Up to degree 3, the odd powers vanish by symmetry, so:

$$P_3(x) = \frac{1}{\sqrt{2\pi}} - \frac{x^2}{2\sqrt{2\pi}}$$

**Theorem 0.14** (Maclaurin polynomial of the standard normal density)**.** *The Maclaurin polynomial (Definition 0.21) of degree 3 of the standard normal density* $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}$ *is*

$$P_3(x) = \frac{1}{\sqrt{2\pi}} - \frac{x^2}{2\sqrt{2\pi}}$$

Proof

*Proof.* This is the solution to Exercise 0.23, worked in Solution 0.23. □

**Exercise 0.24** (Taylor expansion of the square root)**.** Adapted from Miller (2016), Question 1.1.23.
Find the first three terms (up to degree 2) of the Taylor series for

$$f(x) = \sqrt{x}$$

at $x_0 = 1/3$.

Solution

*Solution* 0.24. Compute the values at $x_0 = 1/3$:

$$\begin{aligned} f(1/3) &= \left(\frac{1}{3}\right)^{1/2} \\ &= \frac{1}{\sqrt{3}} \\ &= \frac{\sqrt{3}}{3} \\ f'(x) &= \frac{1}{2}x^{-1/2} \\ \implies f'(1/3) &= \frac{1}{2}\sqrt{3} \\ &= \frac{\sqrt{3}}{2} \\ f''(x) &= -\frac{1}{4}x^{-3/2} \\ \implies f''(1/3) &= -\frac{1}{4}\cdot 3\sqrt{3} \\ &= -\frac{3\sqrt{3}}{4} \end{aligned}$$

The degree-2 Taylor polynomial is:

$$P_2(x) = \frac{\sqrt{3}}{3} + \frac{\sqrt{3}}{2}\left(x - \frac{1}{3}\right) - \frac{3\sqrt{3}}{8}\left(x - \frac{1}{3}\right)^2$$

**Exercise 0.25** (Taylor expansion of a fractional power)**.** Adapted from Miller (2016), Question 1.1.24.
Find the first three terms of the Taylor series for

$$f(x) = (1+x)^{1/3}$$

at $x_0 = 1/2$.

Solution

*Solution* 0.25. At $x_0 = 1/2$, $1 + x_0 = 3/2$:

$$\begin{aligned} f(1/2) &= \left(\frac{3}{2}\right)^{1/3} \\ f'(x) &= \frac{1}{3}(1+x)^{-2/3} \\ \implies f'(1/2) &= \frac{1}{3}\left(\frac{3}{2}\right)^{-2/3} \\ &= \frac{1}{3}\left(\frac{2}{3}\right)^{2/3} \\ f''(x) &= -\frac{2}{9}(1+x)^{-5/3} \\ \implies f''(1/2) &= -\frac{2}{9}\left(\frac{3}{2}\right)^{-5/3} \\ &= -\frac{2}{9}\left(\frac{2}{3}\right)^{5/3} \end{aligned}$$

Dividing $f''(1/2)$ by 2! gives $-\frac{1}{9}\left(\frac{2}{3}\right)^{5/3}$. The Taylor polynomial is:

$$P_2(x) = \left(\frac{3}{2}\right)^{1/3} + \frac{1}{3}\left(\frac{2}{3}\right)^{2/3}\left(x - \frac{1}{2}\right) - \frac{1}{9}\left(\frac{2}{3}\right)^{5/3}\left(x - \frac{1}{2}\right)^2$$

**Exercise 0.26** (Taylor expansion of $x \log x$)**.** Adapted from Miller (2016), Question 1.1.25.
Find the first three terms (up to degree 2) of the Taylor series for

$$f(x) = x \log x$$

at $x_0 = 1$.

Solution

*Solution* 0.26. Evaluate derivatives at $x_0 = 1$:

$$\begin{aligned} f(1) &= 1 \log 1 \\ &= 0 \\ f'(x) &= \log x + 1 \\ \implies f'(1) &= 0 + 1 \\ &= 1 \\ f''(x) = \frac{1}{x} &\implies f''(1) = 1 \end{aligned}$$

Therefore:

$$\begin{aligned} P_2(x) &= f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 \\ &= (x-1) + \frac{1}{2}(x-1)^2 \end{aligned}$$

**Exercise 0.27** (Maclaurin series of $\log(1+x)$)**.** Adapted from Miller (2016), Question 1.1.26.
Find the first three terms of the Taylor series for

$$f(x) = \log(1+x)$$

at $x_0 = 0$.

Solution

*Solution* 0.27. Differentiating repeatedly at $x_0 = 0$:

$$\begin{aligned} f(0) &= \log 1 \\ &= 0 \\ f'(x) = (1+x)^{-1} &\implies f'(0) = 1 \\ f''(x) = -(1+x)^{-2} &\implies f''(0) = -1 \\ f'''(x) = 2(1+x)^{-3} &\implies f'''(0) = 2 \end{aligned}$$

The Taylor polynomial of degree 3 is:

$$P_3(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$$

**Theorem 0.15** (Maclaurin polynomial of $\log(1+x)$)**.** *The Maclaurin polynomial (Definition 0.21) of degree 3 of* $f(x) = \log(1+x)$ *is*

$$P_3(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$$

Proof

*Proof.* This is the solution to Exercise 0.27, worked in Solution 0.27. □

**Exercise 0.28** (Maclaurin series of $\log(1-x)$)**.** Adapted from Miller (2016), Question 1.1.27.
Find the terms up to degree 3 of the Taylor series for

$$f(x) = \log(1-x)$$

at $x_0 = 0$.

Solution

*Solution* 0.28. Substitute $u = -x$ into the expansion for $\log(1+u)$ from Theorem 0.15:

$$\begin{aligned}\log(1-x) &= (-x) - \frac{(-x)^2}{2} + \frac{(-x)^3}{3} - \ldots \\ &= -x - \frac{x^2}{2} - \frac{x^3}{3} - \ldots\end{aligned}$$

**Theorem 0.16** (Maclaurin polynomial of $\log(1-x)$)**.** *The Maclaurin polynomial (Definition 0.21) of degree 3 of* $f(x) = \log(1-x)$ *is*

$$P_3(x) = -x - \frac{x^2}{2} - \frac{x^3}{3}$$

Proof

*Proof.* This is the solution to Exercise 0.28, worked in Solution 0.28. □

**Exercise 0.29** (Taylor series of $\log((1-x)e^x)$)**.** Adapted from Miller (2016), Question 1.1.28.
Find the first two non-zero terms of the Taylor series for

$$f(x) = \log((1-x)e^x)$$

at $x_0 = 0$.

Solution

*Solution* 0.29. Expand using logarithm properties:

$$\begin{aligned} f(x) &= \log(1-x) + \log(e^x) \\ &= \log(1-x) + x \end{aligned}$$

Substitute the series for $\log(1-x)$ from Theorem 0.16:

$$\begin{aligned} f(x) &= \left(-x - \frac{x^2}{2} - \frac{x^3}{3} - ...\right) + x \\ &= -\frac{x^2}{2} - \frac{x^3}{3} - ... \end{aligned}$$

The constant and linear terms vanish ($f(0) = 0$, $f'(0) = 0$). The first two non-zero terms are:

$$-\frac{x^2}{2} - \frac{x^3}{3}$$

**Exercise 0.30** (Taylor series of a product of known expansions)**.** Adapted from Miller (2016), Question 1.1.29.
Find the terms up to degree 2 of the Taylor series for

$$f(x) = \cos(x)\log(1+x)$$

at $x_0 = 0$.

Solution

*Solution* 0.30. Multiply the known Maclaurin expansions:

$$\begin{aligned} \cos x &= 1 - \frac{x^2}{2} + O(x^4) \\ \log(1+x) &= x - \frac{x^2}{2} + O(x^3) \end{aligned}$$

Multiplying and dropping terms of order $x^3$ and higher:

$$
\begin{aligned}
f(x) &= \left(1 - \frac{x^2}{2} + ...\right)\left(x - \frac{x^2}{2} + ...\right) \\
&= x - \frac{x^2}{2} + O(x^3)
\end{aligned}
$$

**Exercise 0.31** (Taylor series of $\log(1 + 2x)$)**.** Adapted from Miller (2016), Question 1.1.30.
Find the first two non-zero terms of the Taylor series for

$$f(x) = \log(1 + 2x)$$

at $x_0 = 0$.

Solution

*Solution* 0.31. Substitute $u = 2x$ into the Maclaurin series $\log(1 + u) = u - \frac{u^2}{2} + ...$:

$$
\begin{aligned}
\log(1 + 2x) &= (2x) - \frac{(2x)^2}{2} + ... \\
&= 2x - 2x^2 + ...
\end{aligned}
$$


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